

One exact seed, with an explicit entropy overhead

Theorem 1. *Let T be a finite controlled tree. There exists a finite seed S , exact for every deterministic policy simultaneously, such that*

$$L \leq H(S) \leq L + \frac{1 + \log_2 e}{2}$$

where L is the mean of the explicit lower envelope constructed in Section 3. In particular

$$H(S) \leq \mathbb{E}Z + \frac{1 + \log_2 e}{2}$$

once Z denotes a random variable with that finite law. No asymptotic statement, Rényi entropy, or Wasserstein distance is used.

The constant is $(1 + \log_2 e)/2 = (1 + 1/\ln 2)/2 = 1.2213475204\dots$. The argument uses only finite sums and one elementary one-dimensional integral. Sections 1–5 are the four steps, in order. Each equality used later is proved where it appears.

1 Model

A *controlled tree* is a finite rooted tree. The root is $\emptyset$. Every nonterminal history v carries a nonempty finite set $\mathcal{A}(v)$ of actions. Every pair (v, A) carries a nonempty finite output alphabet $\mathcal{Y}(v, A)$ and a kernel $K_v(\cdot \mid A)$ with

$$K_v(y \mid A) > 0, \quad \sum_{y \in \mathcal{Y}(v, A)} K_v(y \mid A) = 1.$$

Alphabets may depend on the history and on the action. Zero-probability outputs are deleted, so every kernel entry that appears is positive. The cylinder weights are

$$p(\emptyset) = 1, \quad p(vAy) = p(v) K_v(y \mid A).$$

Thus $p(v) > 0$ at every history that exists, and for each fixed action

$$\sum_y p(vAy) = p(v). \tag{1}$$

A *policy* π chooses, at every history, one action, and specifies a continuation for every output of every action. Along its own action it therefore reaches a finite set $\mathcal{L}_\pi$ of terminal histories. By (1) and induction on the tree,

$$\sum_{v \in \mathcal{L}_\pi} p(v) = 1. \tag{2}$$

The same identity holds for an arbitrary nonnegative flow, not only for p : if r satisfies $r \geq 0$ and $\sum_y r(vAy) = r(v)$ for every history and action, then

$$\sum_{v \in \mathcal{L}_\pi} r(v) = r(\emptyset). \quad (3)$$

A *response strategy* d chooses, at every history and for every action, one output, and specifies a continuation everywhere, including off the histories it will actually visit. A terminal history v is *compatible* with d , written $\chi_d(v) = 1$, when every output on the unique path to v equals the output prescribed by d . The same symbol χ_d is extended to intermediate histories. Exactly one terminal history is compatible both with a given d and with a given π ; call it $\text{run}(d, \pi)$.

A *finite seed* is a finite list of atoms (w_j, d_j) with $w_j > 0$ and $\sum_j w_j = 1$. It is *exact* when, for every policy π and every $v \in \mathcal{L}_\pi$,

$$\sum_{j : \text{run}(d_j, \pi) = v} w_j = p(v). \quad (4)$$

Its Shannon entropy is the ordinary finite sum

$$H(S) = - \sum_j w_j \log_2 w_j.$$

Since $\sum_j w_j = 1$ and $w_j > 0$, one has $w_j \leq 1$, so every logarithm below is defined and nonnegative.

2 Greedy extraction

Let r be a nonnegative flow on the tree. Define B_r from the leaves upward:

$$\begin{aligned} B_r(v) &= r(v) && \text{if } v \text{ is terminal,} \\ B_r(v) &= \min_A \max_y B_r(vAy) && \text{otherwise.} \end{aligned} \quad (5)$$

For a strategy d write $m(d) = \min\{r(v) : \chi_d(v) = 1, v \text{ terminal}\}$. For a policy π write $M(\pi) = \max\{r(v) : v \in \mathcal{L}_\pi\}$.

Lemma 2 (Minimax). *For every nonnegative flow r , and in fact for an arbitrary real assignment of terminal values,*

$$\max_d m(d) = B_r(\emptyset) = \min_\pi M(\pi).$$

Nonnegativity and flow conservation are not used.

Proof. The identity $m(d) = \min_A m(d_{A, y_A})$ is the definition of a strategy: d selects an output y_A for each action and continues as d_{A, y_A} . Proceed by induction on the tree. At a leaf both sides equal r of that leaf, and the only strategy and policy are empty.

At a nonterminal history the inductive hypothesis gives $m(d_A) \leq B_r(vAy_A)$ for the chosen child, and $B_r(vAy_A) \leq \max_y B_r(vAy)$. Hence

$$m(d) = \min_A m(d_A) \leq m(d_{A_*}) \leq B_r(vA_*y_{A_*}) \leq \max_y B_r(vA_*y),$$

where A_* realises the minimum in (5). The last maximum equals $B_r(v)$. Thus every strategy satisfies $m(d) \leq B_r(v)$.

For attainment, choose for each action an output $y_A \in \arg \max_y B_r(vAy)$ and, by the inductive hypothesis, a continuation that attains the child bottleneck. The assembled strategy d_* satisfies

$$m(d_*) = \min_A B_r(vAy_A) = \min_A \max_y B_r(vAy) = B_r(v).$$

Continuations off the chosen outputs may be chosen arbitrarily.

The policy side is dual. A policy selects one action and then every output of that action, so $M(\pi) = \max_y M(\pi_y)$. For every policy, $B_r(v) \leq M(\pi)$, because $B_r(v)$ is at most the action score of the action π selects, and that score is at most $M(\pi)$ by the inductive hypothesis. Choosing a minimising action and an attaining continuation for every output produces a policy with $M(\pi) = B_r(v)$. $\square$

Lemma 3 (Comparison and positivity). *Assume r and s are nonnegative flows and $r \leq s$ at every terminal history. Then $B_r \leq B_s$ at every history. If $r(\emptyset) > 0$, then $B_r(\emptyset) > 0$. If $r(\emptyset) = 0$, then r vanishes at every history. At every history,*

$$B_r(v) \leq r(v). \tag{6}$$

Proof. Pointwise comparison of the terminal data passes through the minimum and the maximum in (5), so $B_r \leq B_s$.

The inequality $B_r(v) \leq r(v)$ is proved from the leaves upward. A leaf is an equality. At a nonterminal history choose any action A_0 . Then $B_r(v) \leq \max_y B_r(vA_0y)$. The inductive hypothesis bounds each child bottleneck by the child mass, and those masses are nonnegative, so a maximum is at most the sum:

$$\max_y B_r(vA_0y) \leq \sum_y r(vA_0y) = r(v).$$

If $r(\emptyset) > 0$, fix any action at the root. Nonnegativity and flow conservation produce a child of positive mass; otherwise every child would vanish and the root would vanish. The inductive hypothesis makes that child's bottleneck positive, so the maximum over outputs is positive. Every action has this property, and the minimum over actions is therefore positive.

If $r(\emptyset) = 0$, nonnegativity and flow conservation force every child of every action to vanish. Induction clears the tree. $\square$

Lemma 4 (One extraction). *Let r be a nonnegative flow with $r(\emptyset) > 0$, let $w = B_r(\emptyset)$, and let d be a strategy with $m(d) = w$. Define $r'(v) = r(v) - w\chi_d(v)$. Then $w > 0$, r' is a nonnegative flow, $r'(\emptyset) = r(\emptyset) - w$, at least one strictly positive terminal value of r becomes zero, and $B_{r'}(\emptyset) \leq w$.*

Proof. Lemma 3 gives $w > 0$. Let v be any history compatible with d . Every alphabet is nonempty and d prescribes an output for every action, so v has a terminal descendant ℓ that is also compatible with d . This terminal is one of the histories whose residual is minimised by $m(d)$, hence

$$r(\ell) \geq m(d) = w.$$

Nonnegativity and flow conservation pass upward along the path from ℓ to v . At each step the path selects one action and one output, and that output is a single nonnegative term in a sum equal to the parent mass, so the parent mass is at least the child mass. Therefore

$$r(v) \geq r(\ell) \geq w,$$

and $r'(v) = r(v) - w \geq 0$. At a history incompatible with d one has $\chi_d(v) = 0$, so the residual is unchanged and remains nonnegative.

Flow conservation of r' holds because, for each action, exactly one output is compatible with d whenever the parent is, and no output is compatible when the parent is not:

$$\sum_y r'(vAy) = \sum_y r(vAy) - w \chi_d(v) = r(v) - w \chi_d(v) = r'(v).$$

The root mass drops by w . The minimum $m(d) = w$ is attained, so some compatible terminal has residual exactly w and becomes zero. Finally $r' \leq r$ at every terminal, so Lemma 3 yields $B_{r'}(\emptyset) \leq w$. $\square$

Proposition 5 (Exact greedy seed). *Starting from $r_0 = p$, iterate Lemma 4. The procedure stops after finitely many steps and returns atoms (w_j, d_j) such that*

1. $w_j > 0$, the sequence (w_j) is nonincreasing, and $\sum_j w_j = 1$;
2. the number of atoms is at most the number of terminal histories of the whole tree;
3. for every history v ,

$$p(v) = \sum_j w_j \chi_{d_j}(v); \tag{7}$$

4. the seed is exact for every policy.

Proof. Each step kills at least one positive terminal, and terminals never become positive again, so the number of steps is at most the number of terminals. Lemma 4 makes the extracted weights nonincreasing. The root mass decreases by w_j at step j and the final root mass is zero: otherwise the bottleneck would still be positive and the iteration would continue. Hence $\sum_j w_j = p(\emptyset) = 1$. The final residual is a nonnegative flow of root mass zero, so it vanishes everywhere by Lemma 3. Telescoping the subtractions gives (7).

For a policy π and a terminal $v \in \mathcal{L}_\pi$, the strategies with $\chi_{d_j}(v) = 1$ are exactly those with $\text{run}(d_j, \pi) = v$. Equation (7) is therefore (4). $\square$

Sampling the atom index with probability w_j , and answering every subsequent query by d_j , is one seed. The decoder does not depend on the policy. Equation (4) says that every policy sees the prescribed transcript law.

3 Explicit envelope

Let $\mathcal{Z}$ be the finite set of all numbers $-\log_2 p(v)$ as π ranges over policies and v ranges over $\mathcal{L}_\pi$. Every such $p(v)$ lies in $(0, 1]$, so every element of $\mathcal{Z}$ is nonnegative and finite. Write

$$z_1 < \cdots < z_m$$

for the ordered elements of $\mathcal{Z}$, and set z_0 formally so that the predecessor of z_1 contributes the value 0 in the difference below. Define

$$F_\pi(t) = \sum_{\substack{v \in \mathcal{L}_\pi \\ -\log_2 p(v) \leq t}} p(v), \quad F_*(t) = \min_\pi F_\pi(t).$$

On each half-open interval $[z_k, z_{k+1})$ every F_π is constant, because no policy has information strictly between z_k and z_{k+1} . Therefore F_* is constant there as well, and

$$q_1 = F_*(z_1), \quad q_k = F_*(z_k) - F_*(z_{k-1}) \quad (k \geq 2). \tag{8}$$

Each F_π is nondecreasing and $F_\pi(z_m) = 1$, so the same is true of F_* . Hence $q_k \geq 0$ and

$$\sum_{k=1}^m q_k = F_*(z_m) = 1. \quad (9)$$

Define

$$L = \sum_{k=1}^m q_k z_k. \quad (10)$$

This is a finite sum. If one prefers a random variable, let Z take the value z_k with probability q_k . Then F_* is the distribution function of Z at every z_k , and $L = \mathbb{E}Z$. No existence argument beyond this finite list is required.

Lemma 6 (Summation by parts). *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be nondecreasing. Then for every policy π ,*

$$\sum_{v \in \mathcal{L}_\pi} p(v) g(-\log_2 p(v)) \leq \sum_{k=1}^m q_k g(z_k). \quad (11)$$

Proof. Let $\mu_k = \sum p(v)$ over the terminals of π with information exactly z_k , and let $S_k = \mu_1 + \dots + \mu_k = F_\pi(z_k)$, with $S_0 = 0$. Then $S_m = 1$ and $S_k \geq F_*(z_k)$. Write $G_k = g(z_k)$, so $G_k \leq G_{k+1}$. Abel's identity is

$$\sum_{k=1}^m \mu_k G_k = \sum_{k=1}^m (S_k - S_{k-1}) G_k = \sum_{k=1}^{m-1} S_k (G_k - G_{k+1}) + S_m G_m.$$

Each difference $G_k - G_{k+1}$ is nonpositive, and $S_k \geq F_*(z_k)$, so replacing S_k by $F_*(z_k)$ increases the sum:

$$\sum_{k=1}^m \mu_k G_k \leq \sum_{k=1}^{m-1} F_*(z_k) (G_k - G_{k+1}) + G_m.$$

The right-hand side reverses to $\sum_{k=1}^m (F_*(z_k) - F_*(z_{k-1})) G_k$, with $F_*(z_0) = 0$. That is $\sum_k q_k g(z_k)$. $\square$

4 The finite majorization

For $\delta > 0$ and $p > 0$ define

$$a_\delta(p) = \begin{cases} p, & \text{if } p \leq \delta, \\ \min\{\delta, p - \delta\}, & \text{if } p > \delta, \end{cases} \quad \psi(u) = \begin{cases} u, & \text{if } 0 \leq u \leq \frac{1}{2}, \\ \frac{1}{2}, & \text{if } \frac{1}{2} < u < 1, \\ 1, & \text{if } u \geq 1. \end{cases} \quad (12)$$

Set $\psi(u) = 0$ for $u < 0$; that case does not occur below. The function ψ is nondecreasing on $\mathbb{R}$: on each piece it is nondecreasing, and the pieces meet, since $\psi(1/2) = 1/2$ and the value immediately to the right of $1/2$ is $1/2$, while ψ approaches $1/2$ from the left of 1 and equals 1 at 1.

Lemma 7 (Pointwise bound). *If $\delta > 0$ and $p > 0$, then $a_\delta(p) \leq p \psi(\delta/p)$. Equality holds for $p \leq \delta$ and for $p \geq 2\delta$.*

Proof. If $p \leq \delta$, then $\delta/p \geq 1$, so $\psi(\delta/p) = 1$ and both sides equal p . If $p \geq 2\delta$, then $\delta/p \leq 1/2$, so $\psi(\delta/p) = \delta/p$ and $p\psi(\delta/p) = \delta$. Also $p - \delta \geq \delta$, so $a_\delta(p) = \delta$. If $\delta < p < 2\delta$, then $1/2 < \delta/p < 1$, so $\psi(\delta/p) = 1/2$ and the right-hand side is $p/2$. Here $p - \delta < \delta$, so $a_\delta(p) = p - \delta$. And $p - \delta \leq p/2$ rearranges to $p \leq 2\delta$, which is true. $\square$

Lemma 8 (Residual mass). *Let (w_j, d_j) be the greedy seed of Proposition 5, and let $\delta > 0$. Write r for the residual flow after every atom with weight strictly larger than δ has been removed. Then*

$$\sum_{w_j \leq \delta} w_j = r(\emptyset), \quad (13)$$

$B_r(\emptyset) \leq \delta$, and there exists a policy π such that

$$r(v) \leq a_\delta(p(v)) \quad \text{for every } v \in \mathcal{L}_\pi. \quad (14)$$

Consequently

$$\sum_{w_j \leq \delta} w_j \leq \sum_{k=1}^m q_k \psi(\delta 2^{z_k}). \quad (15)$$

Proof. The extracted weights are nonincreasing, so the removed atoms are precisely those with $w_j > \delta$, and every remaining atom satisfies $w_j \leq \delta$. The residual root mass is the original mass minus the removed mass, which is (13). If nothing was removed, the first greedy weight is already at most δ , and that weight equals $B_r(\emptyset)$. If something was removed, the residual's bottleneck is the next greedy weight, which was not removed, hence is at most δ . If the residual root vanishes, the bottleneck vanishes as well. In all cases $B_r(\emptyset) \leq \delta$.

By Lemma 2 there is a policy π with $M(\pi) = B_r(\emptyset)$. Every terminal $v \in \mathcal{L}_\pi$ therefore satisfies $r(v) \leq \delta$. Also $0 \leq r(v) \leq p(v)$, because extraction only subtracts. If $p(v) \leq \delta$, then $r(v) \leq p(v) = a_\delta(p(v))$. If $p(v) > \delta$, then $r(v) < p(v)$, so at least one removed atom is compatible with v . Every removed atom has weight $> \delta$, so the subtracted mass at v is $> \delta$ and $r(v) < p(v) - \delta$. Combined with $r(v) \leq \delta$,

$$r(v) \leq \min\{\delta, p(v) - \delta\} = a_\delta(p(v)).$$

The strict inequality $r(v) < p(v) - \delta$ implies the weak inequality required. This is (14).

Flow conservation (3) converts the root mass into a sum along π :

$$\sum_{w_j \leq \delta} w_j = \sum_{v \in \mathcal{L}_\pi} r(v) \leq \sum_{v \in \mathcal{L}_\pi} a_\delta(p(v)).$$

Lemma 7 replaces $a_\delta(p(v))$ by $p(v) \psi(\delta/p(v))$. Since $\delta/p(v) = \delta 2^{-\log_2 p(v)}$,

$$\sum_{w_j \leq \delta} w_j \leq \sum_{v \in \mathcal{L}_\pi} p(v) \psi(\delta 2^{-\log_2 p(v)}).$$

The map $x \mapsto \psi(\delta 2^x)$ is nondecreasing: $\delta 2^x$ is increasing in x and ψ is nondecreasing. Lemma 6 lifts the sum from this single policy to the envelope and gives (15). $\square$

The passage from one policy to the envelope is exactly Lemma 6. It does not choose a new seed, and it does not use a coupling between policies.

5 Entropy

Lemma 9 (Layer cake). *If $w_j > 0$, $\sum_j w_j = 1$, then*

$$H(S) = \int_0^\infty \sum_j w_j \mathbf{1}_{\{w_j \leq 2^{-t}\}} dt. \quad (16)$$

If Z is the finite envelope random variable of Section 3, then

$$L = \int_0^\infty (1 - F_*(t)) dt. \quad (17)$$

Proof. For $0 < w \leq 1$, the condition $w \leq 2^{-t}$ is equivalent to $t \leq -\log_2 w$. The integrand in t therefore equals 1 on a set of length $-\log_2 w$ and equals 0 thereafter. One point does not affect the integral, so

$$\int_0^\infty \mathbf{1}_{\{w \leq 2^{-t}\}} dt = -\log_2 w.$$

Multiply by w_j , sum, and apply the definition of $H(S)$. For the envelope, $Z \geq 0$ takes finitely many values, and the standard layer-cake formula on $[0, \infty)$ reads $\mathbb{E}Z = \int_0^\infty \text{Prob}(Z > t) dt$. But $\text{Prob}(Z > t) = 1 - F_*(t)$. $\square$

Lemma 10 (Integral of the majorant). *For every $z \geq 0$,*

$$\int_0^\infty \psi(2^{z-t}) dt = z + \frac{1}{2} + \frac{1}{2 \ln 2} = z + \frac{1 + \log_2 e}{2}. \quad (18)$$

Proof. Split the integral at $t = z$ and $t = z + 1$.

On $0 \leq t \leq z$, one has $2^{z-t} \geq 1$, so $\psi(2^{z-t}) = 1$. The integral over this range equals z . The value at the single point $t = z$ is immaterial.

On $z < t < z + 1$, one has $1/2 < 2^{z-t} < 1$, so $\psi(2^{z-t}) = 1/2$. At the endpoint $t = z + 1$, one has $2^{z-t} = 1/2$ and $\psi(1/2) = 1/2$. The integral equals $1/2$.

On $t \geq z + 1$, one has $2^{z-t} \leq 1/2$, so $\psi(2^{z-t}) = 2^{z-t}$. Substitute $s = t - z$:

$$\int_{z+1}^\infty 2^{z-t} dt = \int_1^\infty 2^{-s} ds = \int_1^\infty e^{-s \ln 2} ds = \frac{e^{-\ln 2}}{\ln 2} = \frac{1}{2 \ln 2}.$$

Adding the three contributions gives $z + 1/2 + 1/(2 \ln 2)$. Since $\log_2 e = 1/\ln 2$, the constant term equals $(1 + \log_2 e)/2$. $\square$

Lemma 11 (Lower bound for every exact seed). *Let $S = ((w_j, d_j))$ be any finite exact seed, not necessarily the greedy one, and let J be its information, so $\text{Prob}(J \leq t) = \sum_{w_j \geq 2^{-t}} w_j$. Then $\text{Prob}(J \leq t) \leq F_*(t)$ for every t , and therefore $H(S) \geq L$.*

Proof. Fix t and a policy π . Every atom with $w_j \geq 2^{-t}$ is placed by $\text{run}(d_j, \pi)$ into a single terminal $v_j \in \mathcal{L}_\pi$. Exactness gives $w_j \leq p(v_j)$, because w_j is one summand of $p(v_j)$. Hence $p(v_j) \geq 2^{-t}$, that is $-\log_2 p(v_j) \leq t$. Distinct atoms sent to the same terminal contribute separate weights, and the sum of all atom weights sent to terminals of information at most t is at most $\sum_{v: -\log_2 p(v) \leq t} p(v) = F_\pi(t)$. The atoms with $w_j \geq 2^{-t}$ form a subcollection of those atoms. Therefore

$$\text{Prob}(J \leq t) \leq F_\pi(t).$$

The policy was arbitrary, so the left side is at most $F_*(t)$. Equivalently, $\text{Prob}(J > t) \geq 1 - F_*(t)$. Integrate over $t \in [0, \infty)$. Lemma 9 identifies the two sides as $H(S)$ and L , once one observes that $\text{Prob}(J \geq t)$ and $\text{Prob}(J > t)$ differ at only finitely many values of t and have the same integral. The formula in Lemma 9 uses the closed inequality $w_j \leq 2^{-t}$, which is $\text{Prob}(J \geq t)$, and $\text{Prob}(J \geq t) \geq \text{Prob}(J > t)$. $\square$

Proof of Theorem 1. Let S be the greedy seed of Proposition 5. It is exact for every policy. Lemma 11 gives $H(S) \geq L$. For the upper bound apply Lemma 8 at $\delta = 2^{-t}$, for every $t \geq 0$:

$$\sum_j w_j \mathbf{1}_{\{w_j \leq 2^{-t}\}} \leq \sum_{k=1}^m q_k \psi(2^{z_k - t}).$$

Integrate over $t \in [0, \infty)$. The left side is $H(S)$ by Lemma 9. On the right, nonnegativity permits the interchange of a finite sum and the integral:

$$H(S) \leq \sum_{k=1}^m q_k \int_0^\infty \psi(2^{z_k - t}) dt = \sum_{k=1}^m q_k \left(z_k + \frac{1 + \log_2 e}{2} \right) = L + \frac{1 + \log_2 e}{2},$$

where the integral evaluation is Lemma 10 and the last step is (9) together with (10). $\square$

The same integration shows that the inequality is the expectation form $\mathbb{E}J \leq \mathbb{E}[Z + c]$ with $c = (1 + \log_2 e)/2$, but the proof never needs a joint realisation of J and Z . The stochastic sandwich of the longer manuscript is a stronger statement and is not an ingredient here.